



Large Language Models

A Brief Introduction

Large Language Models (LLMs):

sophisticated AI models that can understand and generate human language text to perform various natural language processing (NLP) tasks

How?

LLMs have mastered the art of *Word Guessing*

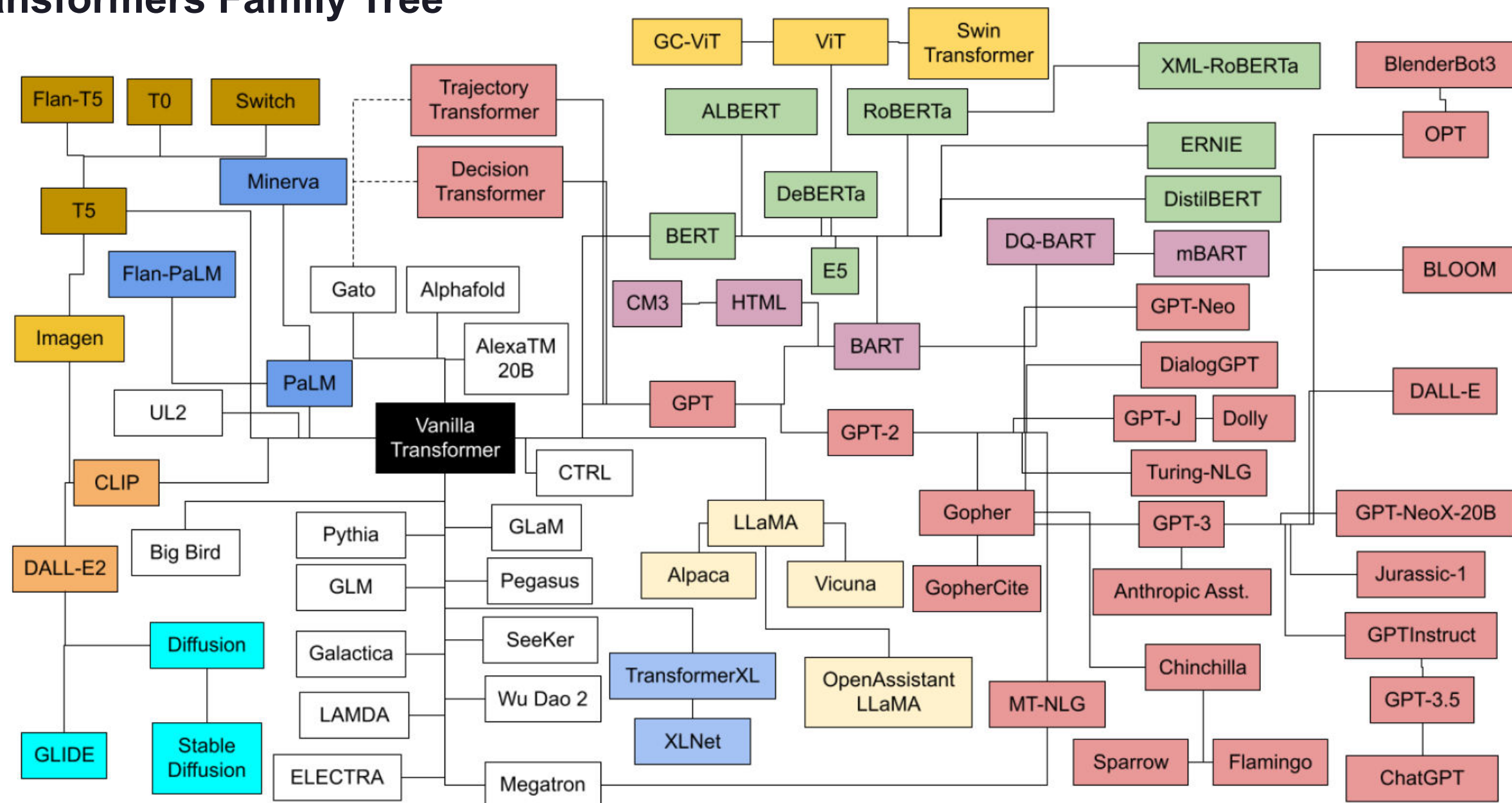
- the ability to forecast the most likely continuation of a given text



LLM Architectures

- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory Networks (LSTMs)
- Gated Recurrent Units (GRUs)
- Convolutional Neural Networks (CNNs)
- *Transformers* ← the dominant architecture today, because of the attention mechanism
 - can process the entire input sequences in parallel, analyze both local and global contexts simultaneously;
 - capable to handle long-range dependencies;
 - can understand relationships and dependencies in data by using self-attention mechanisms

Transformers Family Tree

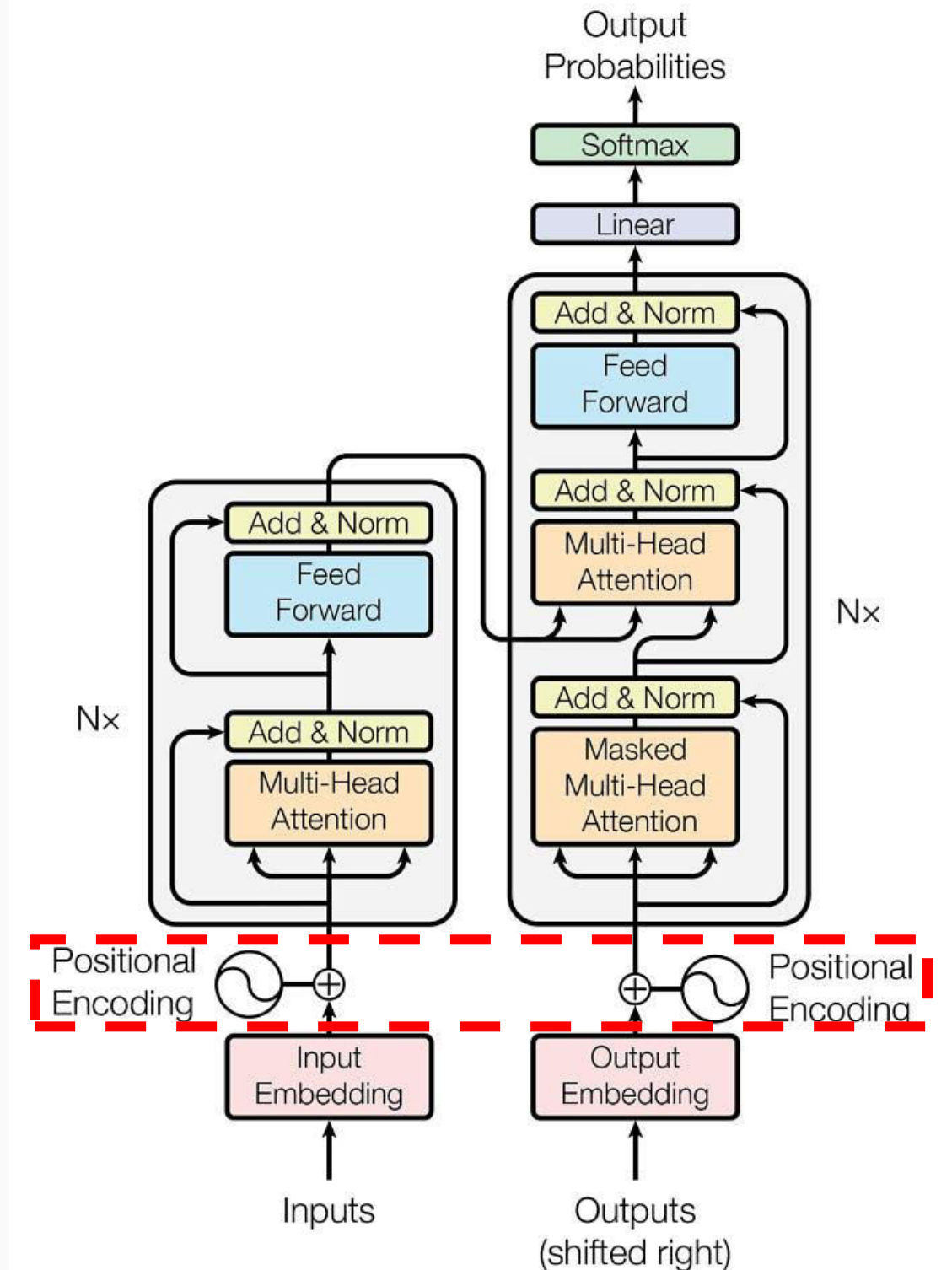
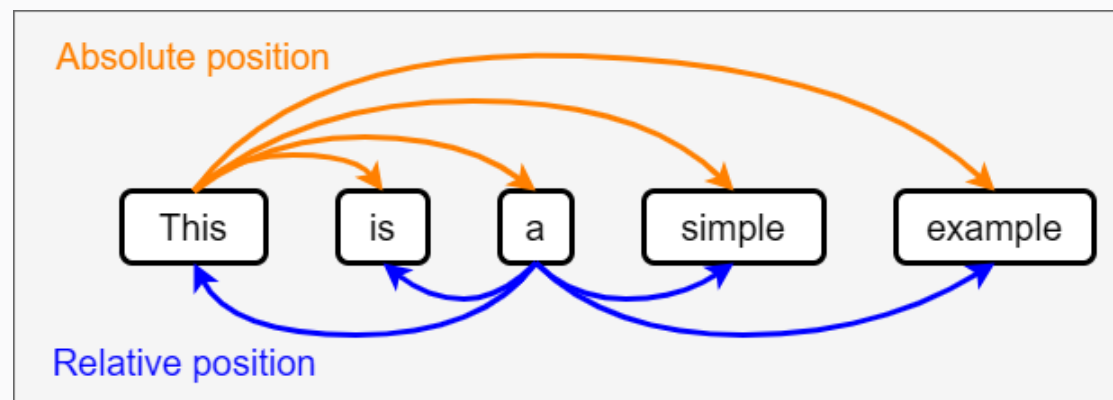


Why Attention Mechanism is used?

- It makes LLMs to **consider the input context** rather than each of the words:
 1. Each word is assigned context-dependent weights based on its relevance to a selected focus
 2. The surrounding words are analyzed to calculate attention scores that represent relevance to the word of interest
 3. The attention scores are then used to weigh the embedding vectors, resulting in **distinct representations of the same word while it carries different contextual meanings** within the input text
- Allows LLMs to capture semantic nuances and improve comprehension, enabling models to selectively focus on relevant parts of input sequences, thereby incorporating context sensitivity into the representation learning process
- Ambiguities can be resolved by attending to informative parts of the input text

Self-attention with Positional Encoding

- Self-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence
- Transformers do not contain recurrence or convolution, positional information of the relative and absolute position of the tokens in the sequence must be added separately
- Positional encodings are added to the input embeddings for both the encoder and decoder



Transformer Models: Encoder-only, Decoder-only, and Encoder-Decoder

Encoder-only models	Decoder-only models	Encoder-Decoder models
• Take the role of analysts to understand the input context; • Extract the information in input context, tokenize it into discrete elements, then transform them into numerical representations (embeddings); • Use case: classification and information extraction	• Like storytellers, generating output text based on the preceding context; • Excel in creative generation of new and coherent text sequences than encoder-only models; • Use case: generative tasks, for example creative writing and question answering	• Leverage the strength of both encoder-only and decoder-only models; • The decoder can focus on relevant information to generate coherent output text; • Use case: text summarization and translation
e.g. BERT, RoBERTa, ELECTRA	e.g. GPT-3, PaLM, LLaMA	e.g. T5, BART

Encoder-only Models

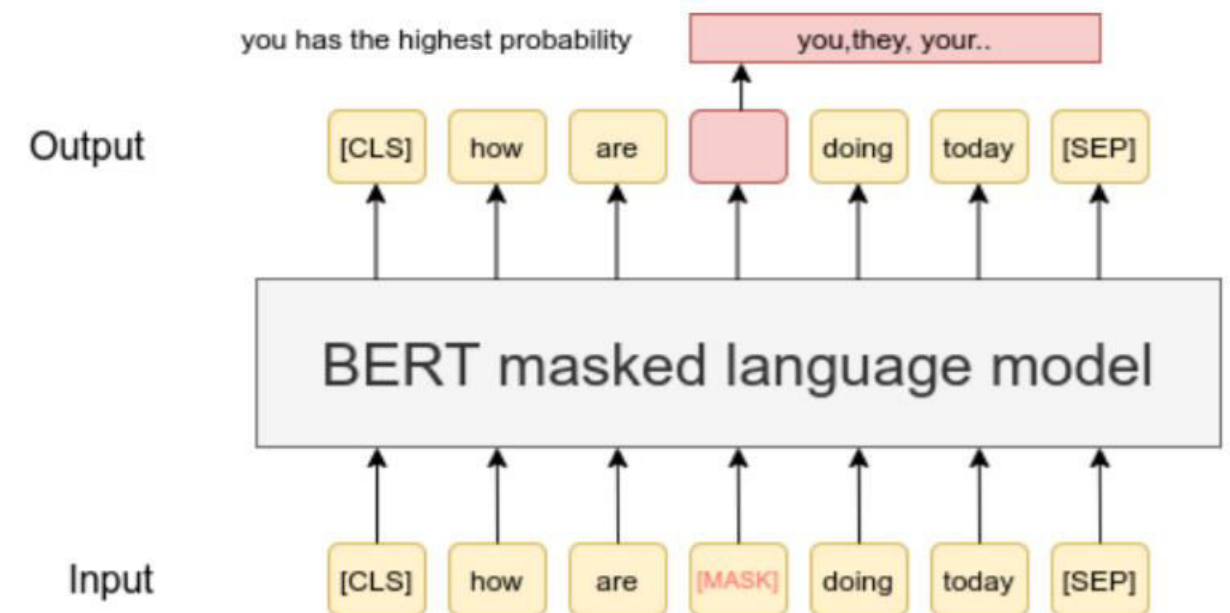
- Autoencoding language models that focus on generating compact vector representations (embeddings) of input text, based on bidirectional context
- Models are pre-trained by masked language modeling (MLM) and next-sentence prediction (NSP) tasks
- MLM: some of the words, e.g. 15%, in the original input text are masked, then let the model predicts the missing words to reconstruct the original input by considering the surrounding context
- NSP: to understand the relationship between two sentences

BIDIRECTIONAL CONTEXT

When predicting words within a sequence, all of the surrounding words can be used to gain contextual information.



Source: <https://towardsdatascience.com/a-complete-guide-to-bert-with-code-9f87602e4a11>



Decoder-only Models

- Autoregressive language models that generate text by predicting the next word in a text sequence based on the previous words
- Pre-training is unidirectional (left-to-right); the future words are masked, so only the previous words can be used to predict the next word
- Probabilistic correlation or dependencies between the current word and previous words is represented by a set of coefficients
- Word prediction is sequential, ensuring the generated text are coherent and contextually relevant

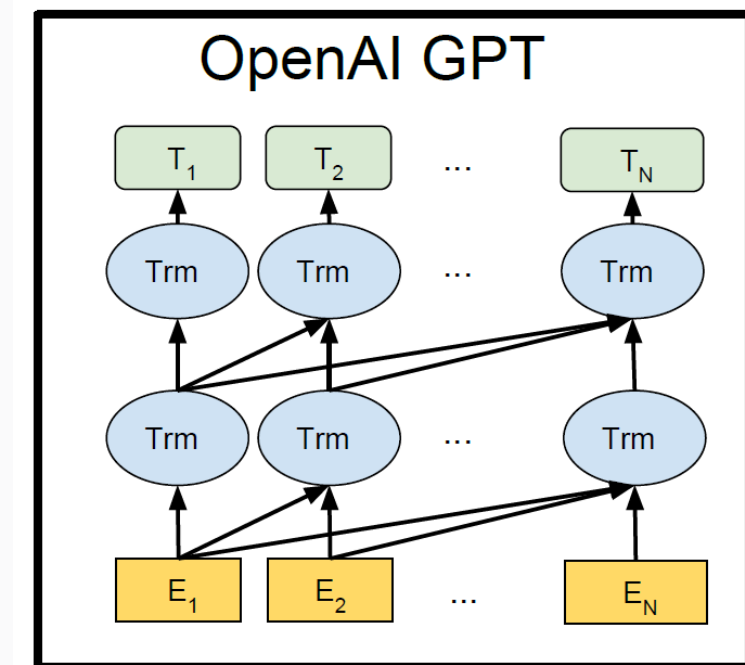
UNIDIRECTIONAL CONTEXT

When predicting future words, only the previous words can be used to gain contextual information.

Left context

Write a poem about a man fishing on a river bank. Upon a [NEXT TOKEN]

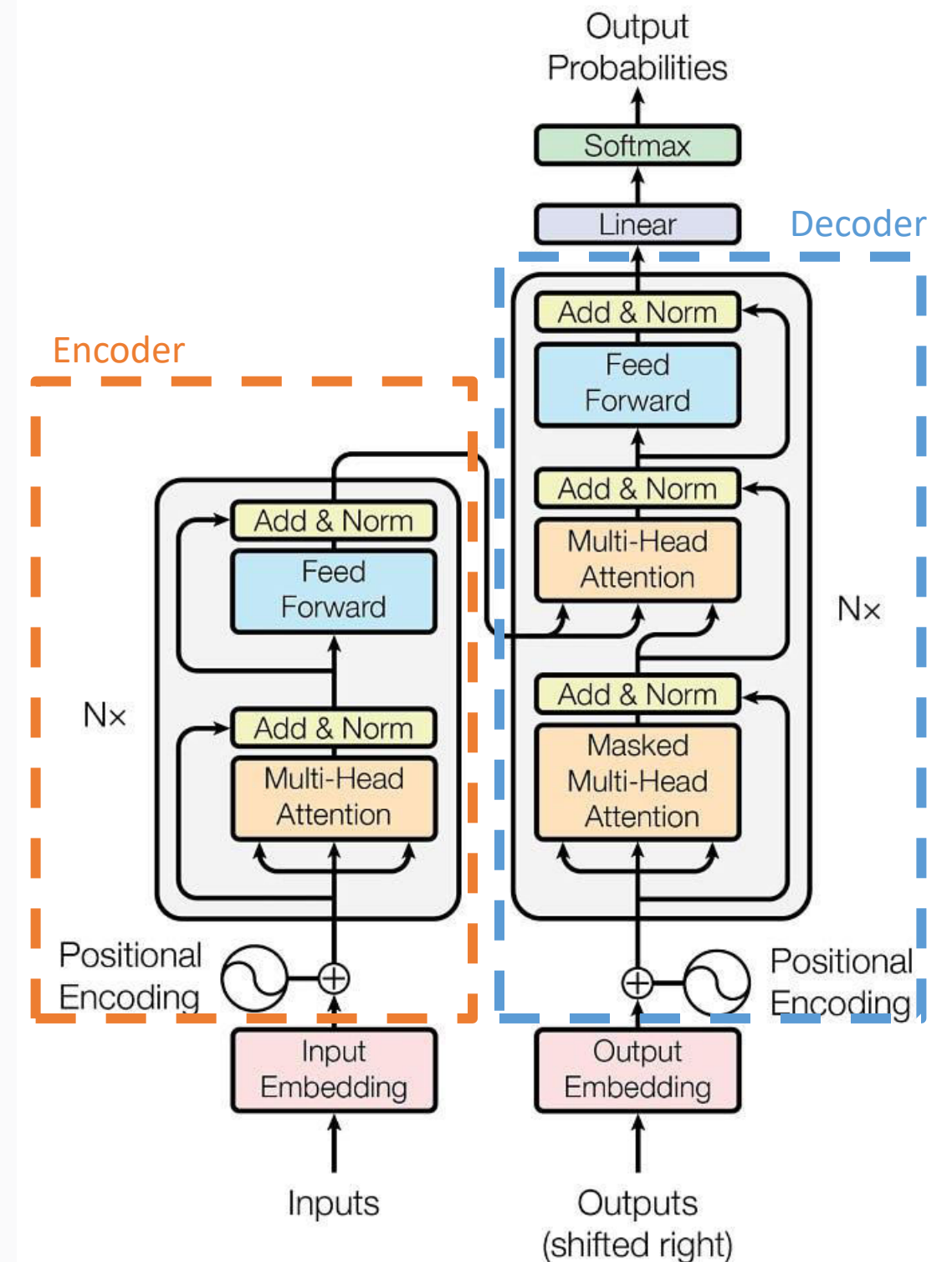
Source: <https://towardsdatascience.com/a-complete-guide-to-bert-with-code-9f87602e4a11>



Davlin et al. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding 2019

Encoder-decoder Models

- Sequence-to-sequence transformer model which consists of an encoder part and a decoder part
- Encoder part receives the input text and then outputs a continuous representation (embedding) of the input text to the decoder
- Decoder generates the output text based on the continuous representation received from the encoder to perform the NLP task
- Particularly good at NLP tasks where there is a complex mapping between the input and output sequences, and where it is crucial to capture the relationships between the elements in both sequences

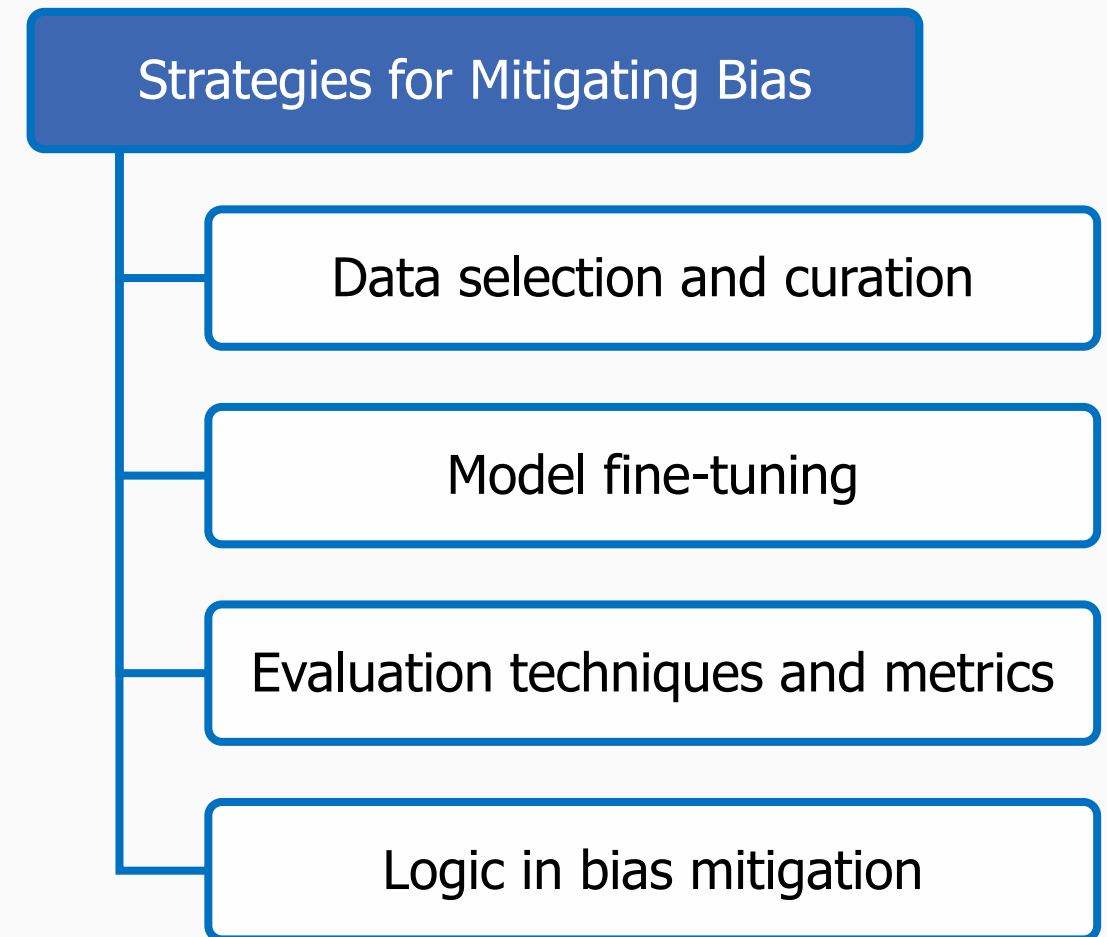


Bias in LLMs

- Bias refers to systematic errors or prejudices in the predictions of LLMs
- Factors causing bias in LLMs:
 1. certain groups or perspectives are misrepresented (can be underrepresented or overrepresented) in the training data;
 2. presence of stereotypes in the sources of training data;
 3. human evaluation such as the design choices and biases of the LLM researchers and developers
- Biased reinforce harmful stereotypes, perpetuate discrimination, amplify existing inequalities in society, and may contribute to the spread of misinformation and disinformation, diminishing trust on the use of AI technologies

Strategies for Mitigating Bias

- Addressing bias in LLMs involves a multifaceted approach that combines technical, ethical, and regulatory measures
- Researchers and practitioners have developed various techniques for detecting and mitigating bias in LLMs, including debiasing algorithms, fairness-aware training procedures, and data augmentation techniques
- Adopting transparent and inclusive practices in data collection, model development, and evaluation can help reduce the risk of bias in LLMs
- Policymakers and regulators should establish guidelines and regulations to promote fairness and accountability in LLMs deployment



Fine-tuning LLMs

- Fine-tuning is a process that adapt a pre-trained LLM for a specific task to refine the behavior of the LLM, to improve its capabilities and precision, to ensure the outputs are consistent and align with the goals
- Depends on the requirements, fine-tuning can be done in different ways:
 1. Full fine-tuning: all the model weights are updated for the specific task, the computation process is intensive
 2. Parameter-efficient fine-tuning (PEFT): the weights of the pretrained LLM remain unchanged. Instead, a small subset of parameters/layers are added to the model and only these parameters are trained
 3. Fine-tuning with reinforcement learning from human feedback (RLHF): use human feedback to train a reward model, and then use the reward model to optimize an agent's policy to align the LLM to human values

Unleashing the Power of LLMs: Insights for Hong Kong Experts

Task Diversity	LLMs can be leveraged for a wide range of language-related tasks, from content generation and translation to question answering and text summarization. Hong Kong experts can explore the versatility of these models and identify specific applications that align with their needs and challenges.
Localization	While LLMs are often trained on a global corpus of data, there is value in fine-tuning and adapting models to the unique linguistic and cultural context of Hong Kong. This can enhance the models' performance and relevance in local applications
Responsible AI	As Hong Kong experts delve deeper into LLMs, it's crucial to address concerns around bias, transparency, and the ethical implications of these powerful technologies. Developing guidelines and best practices for the responsible development and deployment of LLMs will be a key focus.